

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 2.2: Growth and Development in Plants — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 2.2: Growth and Development in Plants
Driving Question	DRIVING QUESTION: What causes the differences in germination, emergence and growth of maize and bean plants?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Meeting the Phenomenon: Mr. Wafula's Maize and Beans	Maize H614 and bean Rosecoco were planted on the same day at Mr. Wafula's farm near Kitale. The bean radicle came out on Day 3 and emerged on Day 8; maize radicle on Day 4 and emerged on Day 7. By Day 14, 88 of 100 maize and 80 of 100 bean seeds had germinated, so 12 and 20 did not, even though some looked healthy. Maize was taller from Day 14 onward, stems thickened, and the bean branched by Day 25.	The two crops behave differently even with the same rain and planting day. Germination (radicle comes out) and emergence (shoot appears above soil) are not the same event. Growth is not only getting taller: stems also thicken and plants branch. We have many questions, such as what growth and development mean and why healthy-looking seeds do not germinate.	Our first ideas are only tentative: we think the differences may be due to the seeds, the soil, water, age or pests, but we do not yet have evidence. Our initial models and DQB questions are our baseline, and we will revise them as we collect evidence. The next question is: What is growth and development?
Lesson 2 Growth vs Development and Inside the Seed: Maize H614 and Rosecoco Beans	We observed soaked Rosecoco bean seeds (testa, hilum, micropyle, two cotyledons, plumule, radicle) and maize H614 grains (one cotyledon called the scutellum, endosperm, coleoptile, coleorhiza, plumule, radicle). We also used Mr. Wafula's data: maize stem width 6, 14, 22 mm and shoot height 3 to 125 cm; bean stem width 2, 4, 6 mm and shoot height 4 to 44 cm.	Growth is a permanent, irreversible increase in size or dry mass; development is a change in complexity and form through differentiation. Monocot (maize) and dicot (bean) seeds have different structures: the bean stores food in two cotyledons, while the maize stores food in the endosperm and one cotyledon (scutellum) absorbs it. Each part has a function, such as protection, food storage, and forming the root and shoot.	Maize and beans differ in seed structure, which is part of why they emerge differently (maize by a coleoptile on Day 7; bean by an arched hypocotyl lifting two cotyledons on Day 8). Both seeds contain an embryo and a testa or coat, but the structure alone does not yet explain why some healthy-looking seeds did not germinate. This raises the next question: What is seed dormancy?
Lesson 3 Seed Dormancy: Why Do Healthy-Looking Seeds Not Germinate?	What we observed: Bean seed testa is hard; scarified seeds took up water faster than untreated ones. In Mr. Wafula's field at Day 14, 12 maize (5 pests/rot, 4 dead	What we learned: Dormancy is when a viable (living) seed fails to germinate even in favourable conditions. Causes include a hard impermeable testa, chemical inhibitors	How this helps explain the driving question: Some healthy-looking bean seeds did not germinate because of a hard, impermeable testa (dormancy), while no maize seeds

	embryos, 3 waterlogged) and 20 beans (8 hard seed coat, 6 waterlogged, 4 dead embryos, 2 pests) did not germinate. 8 dormant beans looked healthy; 7 of the 8 germinated by Day 7 after scarification.	such as abscisic acid, and an immature embryo. Dormancy can be broken by scarification, soaking and other methods that match the cause. Dormancy helps wild seeds survive bad seasons but can delay germination on a farm. Not every failed seed is dormant: some are non-viable, eaten by pests or waterlogged.	were dormant. So differences in germination depend on the seed's structure and condition as well as the environment, not on the rain or planting date. Other failures (waterlogging, lack of oxygen) still need to be explained, which raises our next question: How does germination occur?
Lesson 4 Germination: Conditions Necessary and Types (Hypogeal vs Epigeal) in Maize H614 and Rosecoco Beans	Jar results on Day 5 (20 Rosecoco bean seeds per jar): water + air + 25 C = 18 germinated; dry = 0; no air = 1; cold (4 C) = 0. Time-lapse: bean radicle Day 3, emergence (cotyledons above ground) Day 8; maize radicle Day 4, emergence (coleoptile pierces soil) Day 7, scutellum stays underground. Waterlogged patch: 3 maize and 6 bean seeds failed.	A viable, non-dormant seed needs water, oxygen and a suitable temperature to germinate. Germination is the radicle emerging; emergence is the shoot appearing above the soil. Maize shows hypogeal germination (cotyledon/scutellum stays underground) and beans show epigeal germination (cotyledons lifted above ground by the hypocotyl arch).	Seeds on Mr. Wafula's farm differed because planting on the same day with the same rain does not give every seed the same conditions: the waterlogged patch had no oxygen, so 3 maize and 6 bean seeds failed, while seed structure explains the different emergence days (maize Day 7, bean Day 8). Together with dormancy (Lesson 3), these show that seed condition and environment, not the rain or date, explain differences. This leaves the question: How do plants show growth after emergence?
Lesson 5 Primary and Secondary Growth: Why Does Maize Thicken but Not Make Wood?	We observed sections of maize, bean and Grevillea stems: maize had scattered vascular bundles and no cambium, bean had a ring of bundles with a thin cambium, and Grevillea had a thick ring of wood with annual rings. Maize stem width rose from 6 to 14 to 22 mm; bean rose from 2 to 4 to 6 mm.	Primary growth at apical meristems lengthens shoots and roots. Secondary growth by the vascular cambium thickens woody dicots by adding secondary xylem and phloem and forming annual rings. Maize, a monocot, has no vascular cambium and no true secondary growth.	Maize H614 is wider than Rosecoco bean because its stem thickens through primary growth, not because it makes wood. Only woody dicots like Grevillea thicken by secondary growth. This adds to the explanation of differences in growth between the two crops on Mr. Wafula's farm. New question: How can you measure growth in plants?
Lesson 6 Measuring Growth: Regions of Growth and Growth Curves of Maize H614 and Bean Rosecoco	Marks placed on maize and bean root tips spread apart most just behind the tip, showing growth occurs in a limited region. The height data for Days 7 to 42 showed maize H614 rising from 3 to 125 cm while bean Rosecoco rose from 4 to 44 cm. Maize increased most between Day 28 and 35 (36 cm) and bean between Day 21 and 28 (12 cm) in the table. Root length at Day 42 was 40 cm for maize and 26 cm for bean.	Growth can be measured as height, stem width or root length. Growth in roots and shoots occurs at the apical meristem (cell division), followed by a region of elongation and a region of differentiation. A growth curve is sigmoid, with lag, log and stationary phases. Maize grows faster and for longer than beans, with its steepest growth between Day 21 and 35, while bean is steepest between Day 14 and 28 and then flattens.	Maize and bean plants differ in growth, not only in germination and emergence, because they grow at different rates and for different lengths of time, even though they were planted on the same day in the same field. The curves show when growth is fastest and when it slows, but they do not explain what controls it. This raises the question for Lesson 7: Do we have hormones in plants?
Lesson 7 Plant Hormones and Apical Dominance: What	We watched the video on major plant hormones and read about their roles. We looked at Mr. Wafula's bean data: with the	Plant hormones coordinate growth and development. Auxin (IAA) is made in the shoot tip and promotes elongation while	The bean with its tip intact did not branch because auxin from the apical meristem inhibited the lateral buds. Removing the tip

Coordinates Growth in Maize and Beans?	shoot tip intact, 1 main stem and 0 branches at Day 25; with the tip removed at Day 20, 3 lateral branches by Day 32. We also recalled that 7 of the 8 dormant bean seeds germinated by Day 7 after scarification.	inhibiting lateral buds (apical dominance). Cytokinins promote cell division and lateral bud growth. Gibberellins promote germination and stem elongation. Absciscic acid maintains dormancy. Ethylene promotes fruit ripening.	lowered auxin, so cytokinin could promote lateral buds and 3 branches grew by Day 32. Hormones such as abscisic acid and gibberellins help explain dormancy and germination, so the differences between maize H614 and bean Rosecoco depend on seed structure, seed condition, environment and hormone coordination of growth.
Lesson 8 Applying Hormones: Rooting Cuttings for the School Tree Nursery	We observed that hibiscus or Duranta cuttings treated with natural sources rooted better than cuttings in plain water: out of 30 cuttings, plain water 9 (30%), bulb onion 21 (70%), Aloe vera gel 24 (80%) and cinnamon powder 18 (60%).	We learned that a fair test changes only one variable and keeps others the same, that percentages let us compare treatments, and that hormones such as auxin at the cut base help cuttings form roots, so hormones can be applied in a school tree nursery.	Aloe vera gel is the best treatment because 80% of its cuttings rooted compared with 30% in the control, and the only difference was the treatment. This shows hormones affect plant growth and development, which helps answer our driving question about why plants grow differently: seed structure, seed condition, environment and hormones all play a part.
Lesson 9 Consolidating the Evidence and Building the Final Model: Why Did Mr. Wafula's Maize and Beans Behave So Differently?	We observed that on Mr. Wafula's farm near Kitale, maize H614 (88 of 100 germinated by Day 14, emerged Day 7, 125 cm by Day 42) and Rosecoco beans (80 of 100 germinated, emerged Day 8, 44 cm by Day 42) behaved differently even though they were planted on the same day after 25 mm of rain. Non-germinating seeds: maize 12 (5 pests/fungi, 4 non-viable, 3 waterlogged) and beans 20 (8 hard seed coat, 6 waterlogged, 4 non-viable, 2 pests). After scarification, 7 of the 8 dormant beans germinated.	We learned that the seeds differ in structure (maize is a monocot and hypogeal; bean is a dicot and epigeal), that germination needs water, oxygen and a suitable temperature, that dormancy means a viable seed fails to germinate even in good conditions, that primary growth at meristems gives a sigmoid growth curve, that maize has no vascular cambium and no true secondary growth, and that hormones (auxin, gibberellins, cytokinins, abscisic acid) coordinate growth, including apical dominance.	The differences came from seed structure, seed condition (dormancy, viability, pests) and the environment (such as the waterlogged patch), not from the rain or the planting date. Once growing, hormones coordinate growth: the maize grows taller and thicker through primary growth, while the bean branches only when the tip is removed, because auxin from the apical bud inhibits the lateral buds.
Lesson 10 Presenting Our Final Explanation and Sharing Seedlings with Kitale	We observed our own Day 1 models compared with our final models, the evidence on other groups' posters (germination 88 maize vs 80 beans out of 100; 8 dormant beans, 7 germinating after scarification; emergence Day 7 for maize and Day 8 for beans; heights at Day 42 of 125 cm for maize and 44 cm for beans; 0 branches vs 3 branches after removing the tip), and the nursery seedling sales plan (60 x KES 20 = KES 1,200, cost KES 300, profit KES 900).	We learned that the differences were not caused by the rain or planting date. They come from seed structure (maize monocot and hypogeal; beans dicot and epigeal), seed condition (dormancy from a hard seed coat, non-viable seeds, pests) and environment (waterlogging and lack of oxygen), and that hormones (auxin, gibberellins, cytokinins, abscisic acid) coordinate primary growth and branching. We also learned that our science can be used to grow and sell tree seedlings for conservation and income, as Dr. Wangari Maathai inspired.	Claim: The maize and beans on Mr. Wafula's farm differ because of seed structure, seed condition and environment, not because of rain or planting date. Evidence: Day 14 germination of 88 maize vs 80 beans; the failures were pests, non-viable embryos, waterlogging and, in beans, 8 dormant seeds, 7 of which germinated after scarification; maize emerged on Day 7 and beans on Day 8; Day 42 heights were 125 cm and 44 cm; beans stayed unbranched with the tip but formed 3 branches after the tip was removed. Reasoning: structure decides how seedlings emerge, a hard testa blocks

			water, no oxygen stops respiration, meristems and hormones coordinate growth, and apical dominance explains branching, so farmers can improve yields and communities can grow trees.
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